

DIAGNOSIS OF THE MANUFACTURING INDUSTRY IN BOLIVIA

Context of the Manufacturing Industry

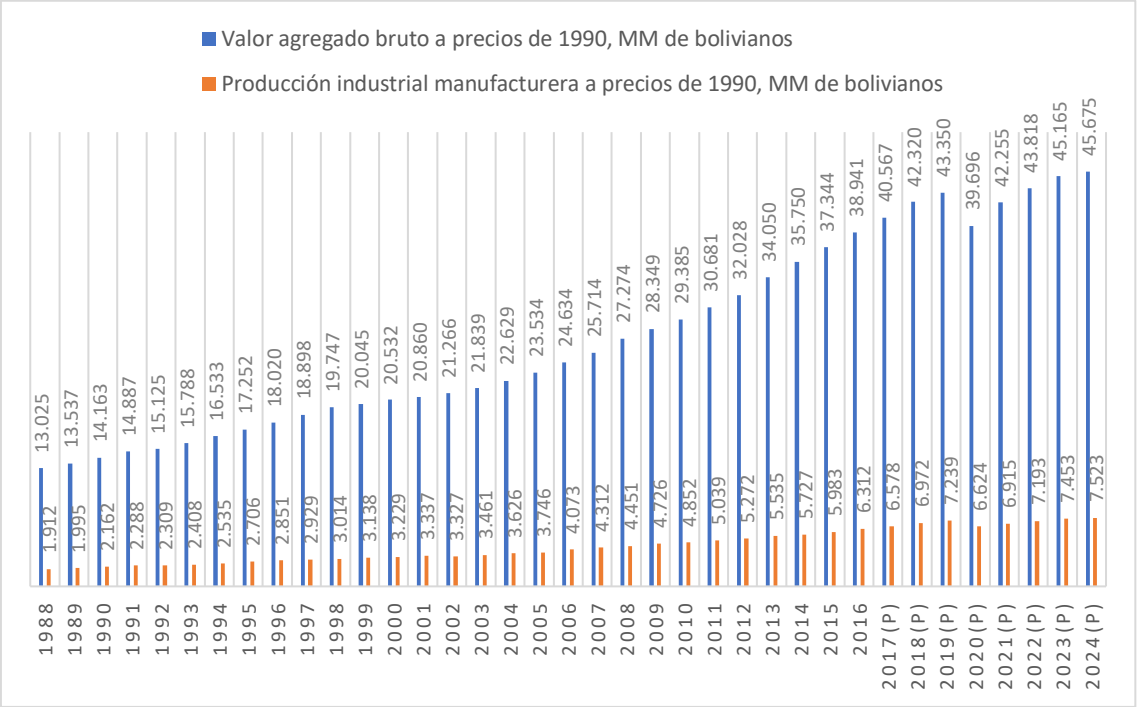
Activities related to industrial manufacturing, excluding those associated with mining-metallurgical activities and petroleum refining, are included in the following sectors measured in the Bolivian economy through input-output matrices:

- 8. FRESH AND PROCESSED MEAT
- 9. DAIRY PRODUCTS
- 10. MILLING AND BAKERY PRODUCTS
- 11. SUGAR AND CONFECTIONERY
- 12. MISCELLANEOUS FOOD PRODUCTS
- 13. BEVERAGES
- 14. PROCESSED TOBACCO
- 15. TEXTILES, APPAREL, AND LEATHER PRODUCTS
- 16. WOOD AND WOOD PRODUCTS
- 17. PAPER AND PAPER PRODUCTS
- 18. CHEMICAL SUBSTANCES AND PRODUCTS
- 20. NON-METALLIC MINERAL PRODUCTS (glass, cement)
- 21. BASIC METAL PRODUCTS
- 22. METAL PRODUCTS, MACHINERY, AND EQUIPMENT
- 23. MISCELLANEOUS MANUFACTURED PRODUCTS

The evolution of the value added of these sectors over the last 40 years is shown in the following figure, compared with the evolution of GDP, expressed in millions of 1990 bolivianos.

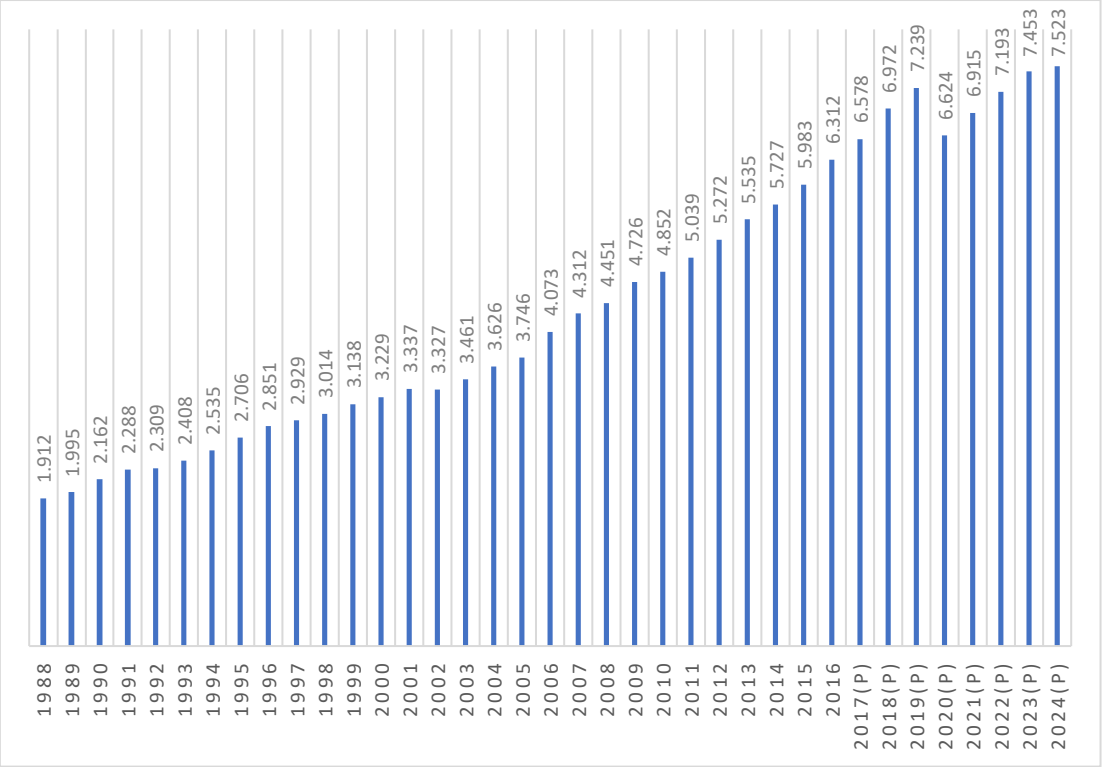
Likewise, Figure 2 shows the evolution of value added corresponding to the manufacturing sectors indicated above, used as a proxy for the gross production value of this sector, considering that this information is currently limited in the data published by the National Institute of Statistics of Bolivia.

Figure 1. Evolution of manufacturing industrial production under the MDPyEP, compared with GDP, in millions of 1990 Bs.



Source: Prepared by the author based on INE data.

Figure 2. Evolution of industrial production in manufacturing sectors, in millions of 1990 Bs.

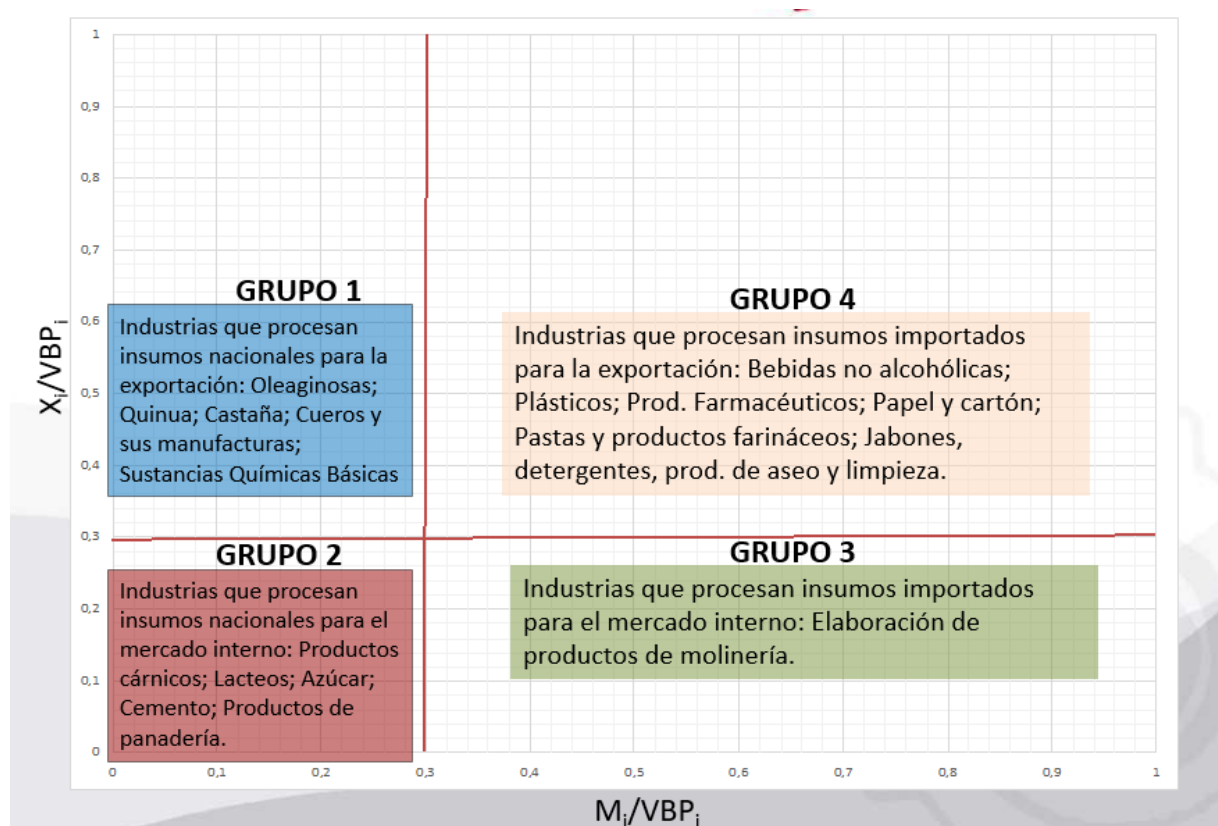


Source: Prepared by the author based on INE data.

The industrial production indicated above has shown sustained growth over the last 40 years, with a decline in 2020, a period corresponding to the COVID-19 pandemic and the de facto government, followed by a recovery from 2021 onward.

Regarding the structure of the sector, the following chart, prepared in 2015 by former Vice Minister of Internal Trade and Exports Luis Baudoin, assuming that the structure has not changed significantly, presents a classification of industrial activities according to their dependence on imported inputs and their exports, distinguishing four areas as shown in the following figure.

Figure 3. Structure of the industrial sector



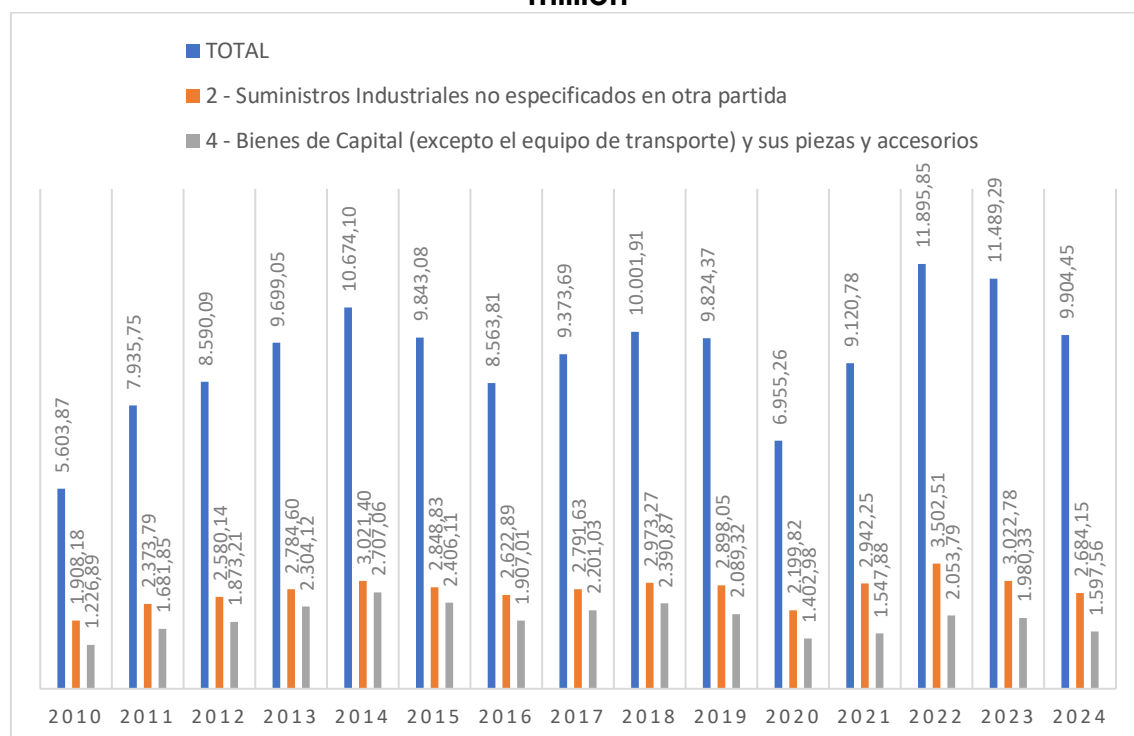
Source: Documents of the Vice Ministry of Internal Trade and Exports, 2015.

It can be observed that there are very important and highly dynamic sectors, such as non-alcoholic beverages, plastics, paper, pharmaceutical products, and detergents, which depend on imported inputs (Group 3). Meanwhile, other sectors are oriented toward the domestic market and use national inputs; this is the sector that should grow through stronger productive linkages and by exporting the surpluses generated, as is the case with much of food production (the food industry, including meat products, dairy products, sugar, and cement, among the most important). It is also desirable for this structure to incorporate industrial

activities that include technology and greater value added, which represents a challenge for industrial development.

It should be emphasized that the manufacturing industrial sector is highly dependent on imports of capital goods and inputs, as shown in the following graphs, whose amounts are expressed in millions of dollars.

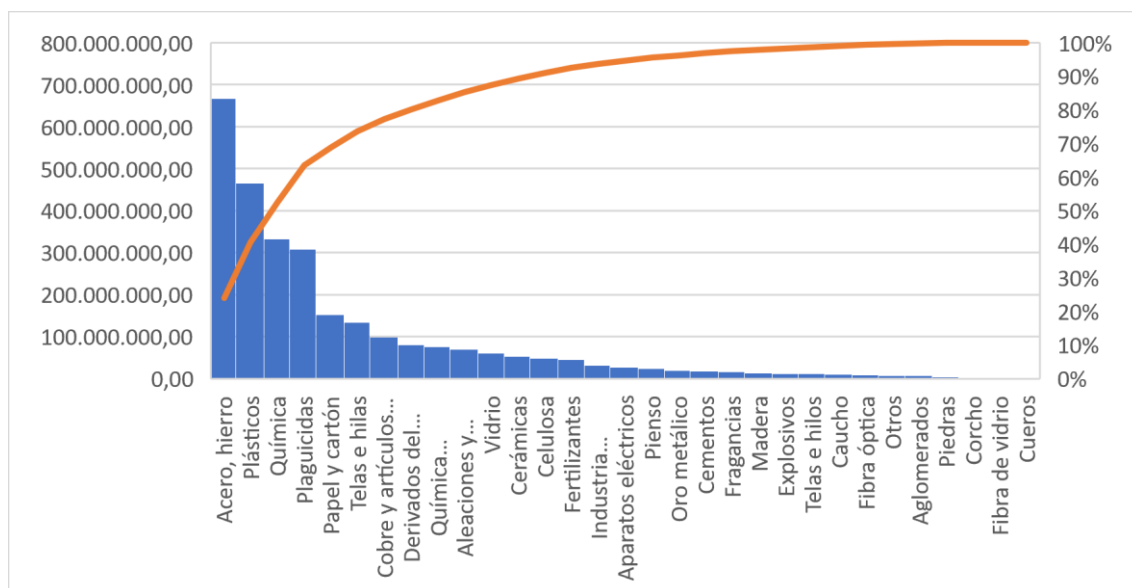
Figure 4. Imports highlighting industrial supplies and capital goods, in USD million



Source: Prepared by the author based on INE data.

Regarding imports of inputs for industry, the highlighted imports related to manufacturing include fuels, which represent approximately USD 2.8 billion per year and are cross-cutting for industrial activity; agricultural pesticides and fungicides, approximately USD 300 million per year, which are related to food production, with the food industry being one of the most important; and chemical inputs for both industry in general and the pharmaceutical industry, which together represent more than USD 800 million per year. Moreover, as in the case of basic chemicals, dependence on these imports limits the development of this industrial subsector in the country, which is key for various industries such as metal treatment, inputs for the detergent industry, water treatment, food additives, glass production, cement production, and the lithium industry. The following figure shows the import of industrial inputs, considering the main categories, as an average of recent years, with data expressed in United States dollars.

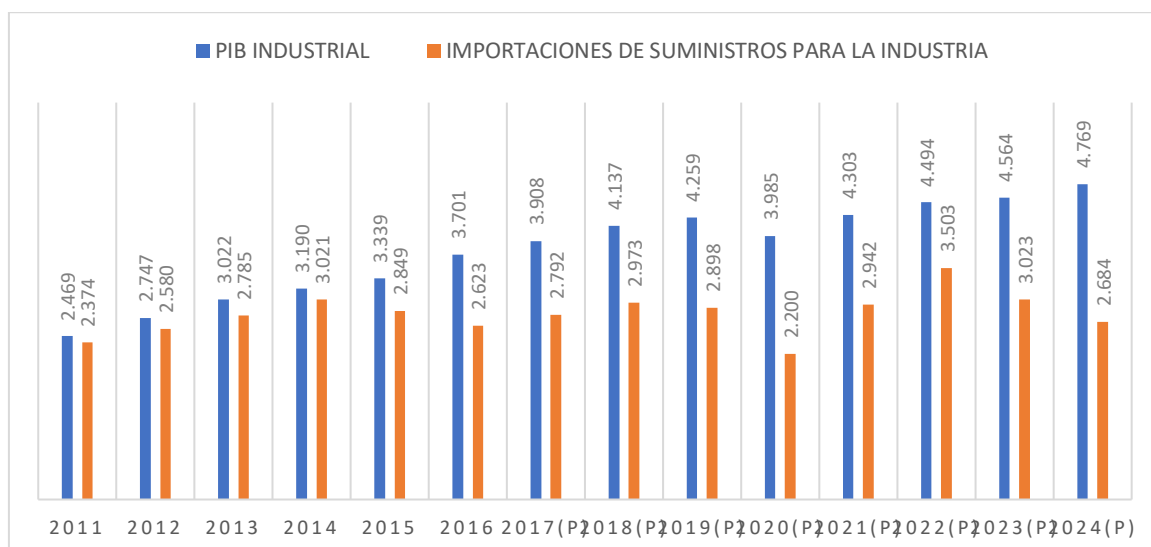
Figure 5. Imports of the main industrial inputs for Bolivian industry, in United States dollars



Source: Prepared by the author based on INE data.

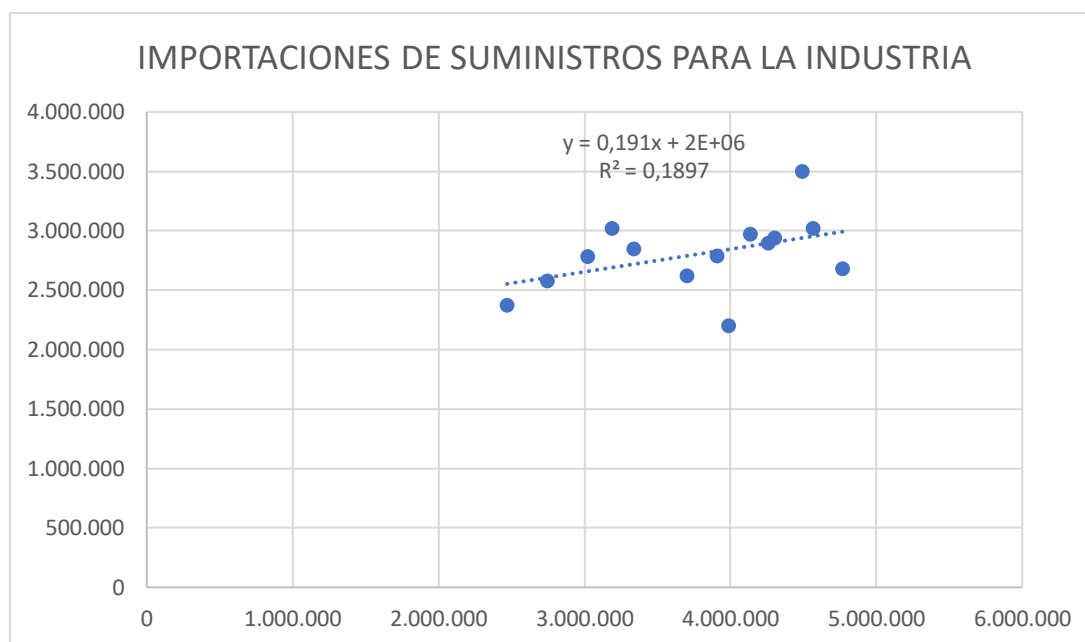
Likewise, the following figures show the positive relationship between industrial value added and the import of inputs, including trend lines.

Figure 6. Value added of the manufacturing sector compared with input imports, both in United States dollars



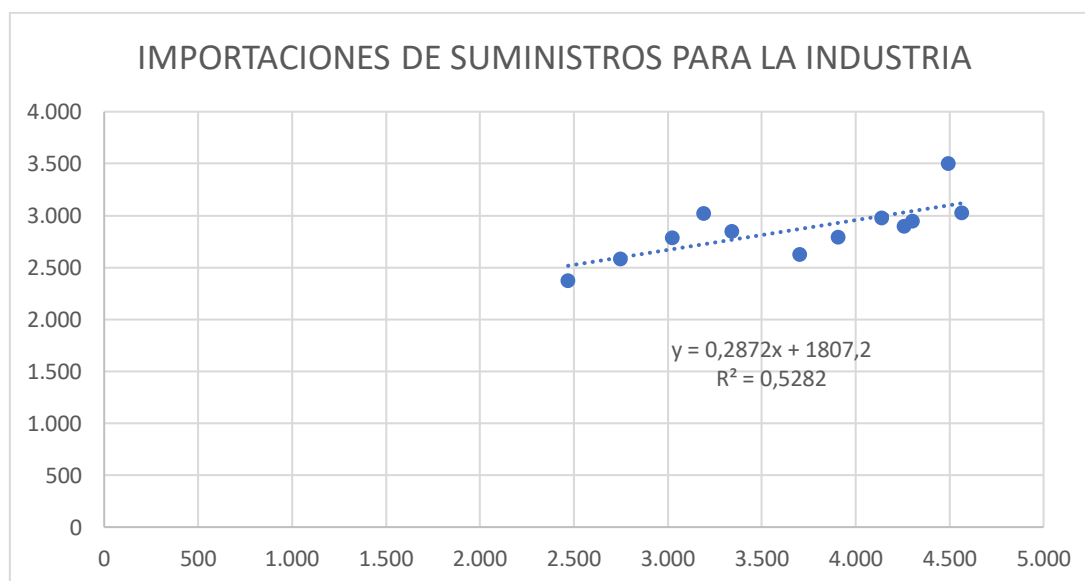
Source: Prepared by the author based on INE data.

Figure 7. Relationship between industrial value added and input imports



Source: Prepared by the author based on INE data.

Figure 8. Relationship between industrial value added and input imports, excluding three points that deviate significantly from the trend line



Source: Prepared by the author based on INE data.

A positive correlation is observed among the plotted data, showing that the increase in Bolivia's industrial production has been accompanied by imports of industrial inputs.

If a regression analysis is performed, considering industrial production as the independent variable and imports as the dependent variable, and excluding the observations corresponding to 2019 and 2024 because they

show atypical behavior, the resulting data are presented in the following table:

Table 1. Regression parameters between the value added of industrial production and imports of industrial supplies

<i>Regression statistics</i>				
Multiple correlation coefficient	0,77084374			
Coefficient of determination R ²	0,59420008			
R ² ajustado	0,54911119			
Standard error	197,04554			

	<i>Coefficients</i>	<i>Standard error</i>	<i>t-statistic</i>	<i>Probability</i>
Intercept	1717,3403	323,226223	5,31312184	0,00048537
Variable X 1	0,3071157	0,08460001	3,63020896	0,00548384

Source: Prepared by the author.

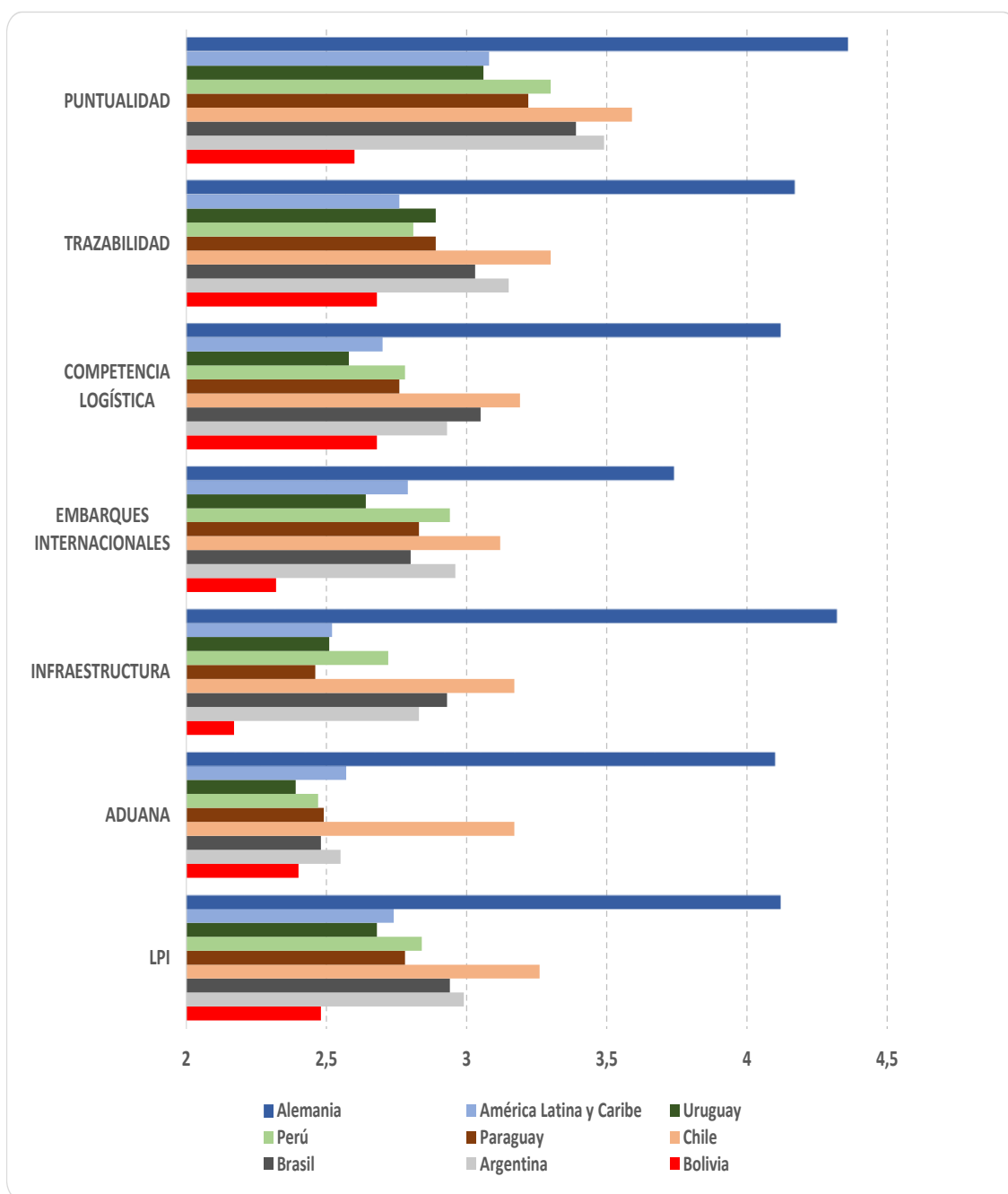
The coefficient of the independent variable is observed to be significant at the 0.5% level.

Logistical limitations

With respect to connectivity, it is necessary to refer to the serious logistical problems faced by the country, not only in relation to its landlocked condition but also to its limited infrastructure and information management systems. Nevertheless, through an intelligent integration policy, Bolivia could take advantage of its geographical position in connection with major projects that are no longer under discussion, such as the bioceanic railway, the development of river transport, dry ports, and logistics infrastructure. This would also require strengthening the use of information management technologies and capitalizing on the opportunities represented by Brazil's need for greater connection with the Pacific and, therefore, with China, as well as Peru's port development, among other factors.

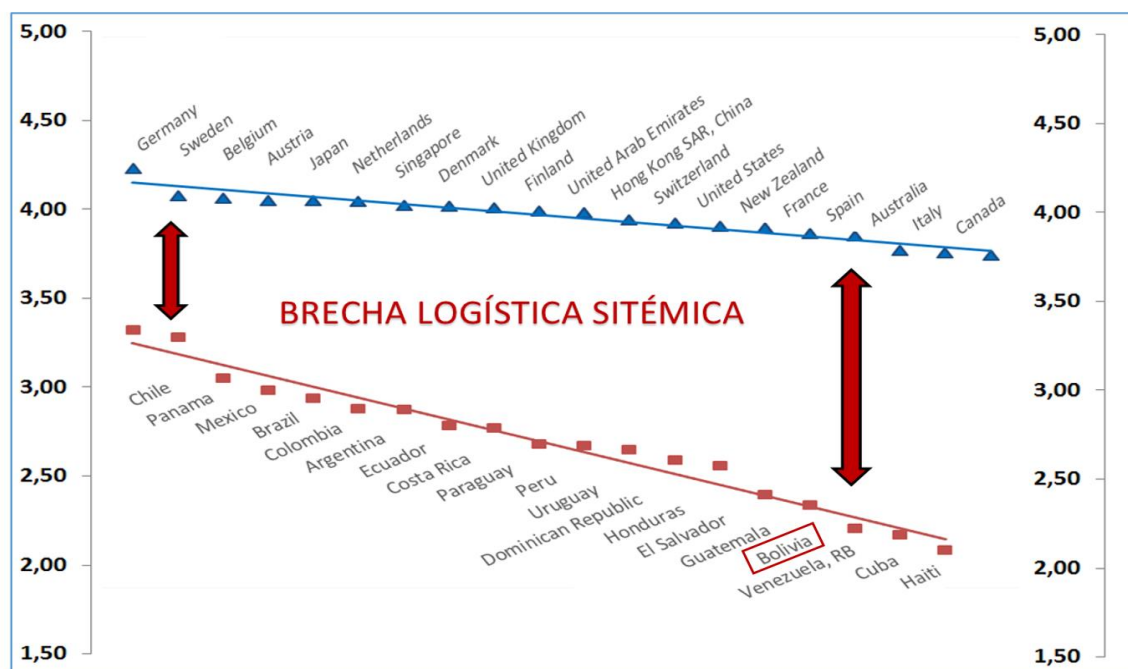
The figures presented below come from a diagnosis of Bolivia's logistics situation prepared in 2015, using information from 2006 and 2014, by the technical team of the Export Directorate of the Ministry of Productive Development and Plural Economy, under the leadership of Mr. Marcelo Olguin. The diagnosis presents the Logistics Performance Index and its comparison with other countries and regions. It is considered that the current logistics situation has not changed significantly.

Figure 9. Logistics performance of Bolivia



Source: Ministry of Productive Development and Plural Economy, VCIE, DGE, 2015.

Figure 10. Comparison of logistics performance with other countries and regions



Source: Ministry of Productive Development and Plural Economy, VCIE, DGE, 2015.

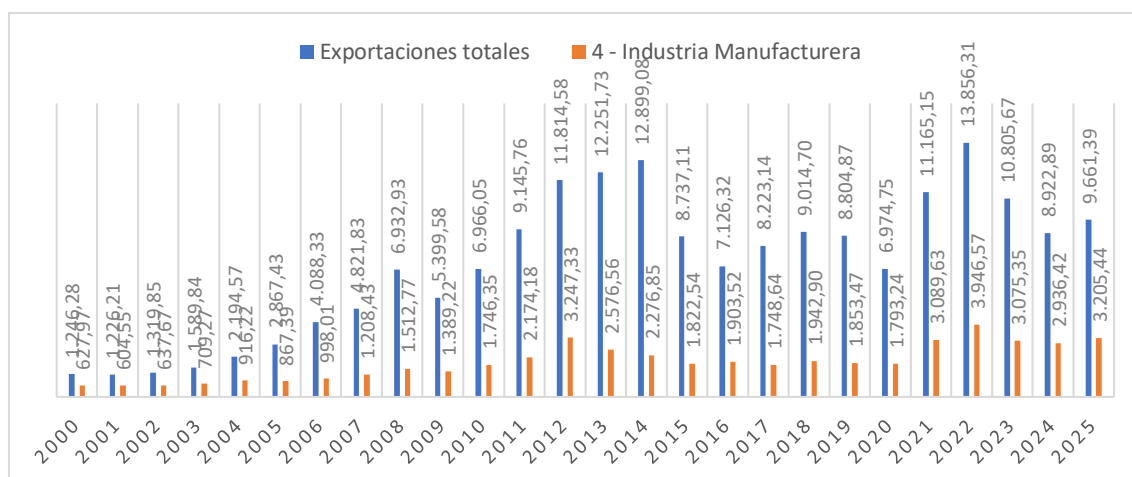
Likewise, the National Logistics Survey carried out in 2019 with the support of the Inter-American Development Bank determined that approximately 20% of the cost of imported and/or exported goods is associated with transport and logistics, a figure well above that of other countries in the region.

Finally, it is necessary to comment on weak international integration and the chaotic or absent policy for developing trade instruments that, under a strategic vision, should enable these mechanisms to be used intelligently to position Bolivian value-added products in market niches. This should take advantage of unique and unrepeatable conditions such as royal quinoa, llama fiber and meat, high-altitude wine, and others, as well as the development of products incorporating technology, such as robotics and software development.

Exports

Regarding manufacturing exports, the following figure shows their evolution compared with the country's total exports, showing that, over time, they have represented less than 60% of Bolivian exports.

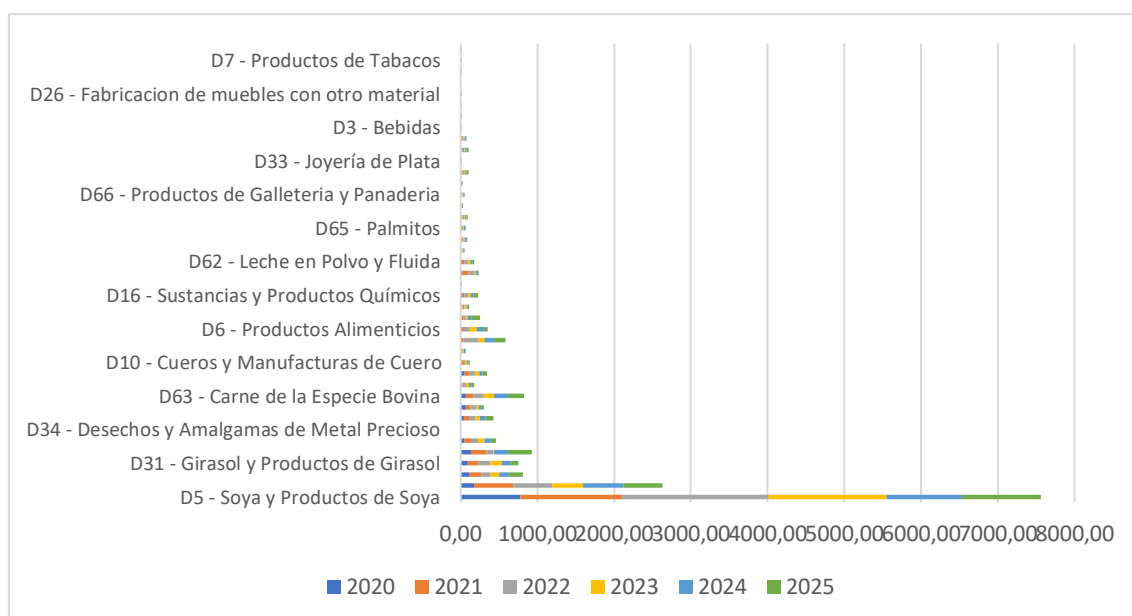
Figure 11. Manufacturing exports compared with total exports, in millions of dollars



Source: Prepared by the author based on INE data.

Regarding the composition of Bolivian exports, the following figure shows the situation for the last five years, in millions of dollars.

Figure 12. Manufacturing exports by category, in millions of dollars



As can be observed, Bolivian manufacturing exports are composed of basic products with low value added, mostly from the agricultural sector. Among processed products, most correspond to food, while exports of products incorporating advanced technology are almost nonexistent.

Analysis of Forward and Backward Multipliers

This section presents an analysis of the input-output matrix using the Leontief model, based on the latest matrix published by INE (2014). The calculations indicated below are carried out in order to determine the key

sectors of the economy and how productive investment should be prioritized.

The calculations follow the logic shown in the figure below.

Figure 13. Leontief model using the input-output matrix

$$X = (I - A)^{-1} \cdot y$$

$$(I - A)^{-1} = \begin{pmatrix} 1,83 & 0,64 & 0,48 \\ 1,35 & 1,86 & 0,50 \\ 1,66 & 1,66 & 1,66 \end{pmatrix}$$

Multiplicado hacia atras

Multiplicadores hacia adelante

Source: The Leontief Matrix, Waldo Vasquez Gonzales.

The calculations are applied to the 2015 input-output matrix, the latest one published by INE Bolivia at the time this work was prepared, assuming that the structure of the economy has not changed significantly.

The indicated INPUT-OUTPUT matrix is presented below.

Table 2. Input-output matrix, 2015

ACTIVIDAD ECONÓMICA		1. PRODUCTOS AGRÍCOLAS NO INDUSTRIALES	2. PRODUCTOS AGRÍCOLAS INDUSTRIALES	3. COCA	4. PRODUCTOS PECUARIOS	5. SILVICULTURA, CAZA Y PESCA	6. PETRÓLEO CRUDO Y GAS NATURAL	7. MINERALES METÁLICOS Y NO METÁLICOS	8. CARNES FRESCAS Y ELABORADAS	9. PRODUCTOS LÁCTEOS	10. PRODUCTOS DE PANADERÍA Y CONFITERÍA	11. AZÚCAR Y PRODUCTOS ALIMENTICIOS DIVERSOS	12. BEBIDAS	13. TABACO ELABORADO	14. TEXTILES, PRENDAS DE VESTIR Y PRODUCTOS DEL CUERO	15. MADERA Y PRODUCTOS DE MADERA	16. PAPEL Y PRODUCTOS DE PAPEL	17. SUBSTANCIAS QUÍMICAS Y PRODUCTOS QUÍMICOS	18. PRODUCTOS DE REFINACIÓN DEL PETRÓLEO	19. PRODUCTOS BÁSICOS DE METALES	20. PRODUCTOS METÁLICOS, MAQUINARI Y EQUIPO	21. PRODUCTOS MANUFACTURADOS DIVERSOS	22. ELECTRICIDAD, GAS Y AGUA	23. CONSTRUCCIÓN	24. COMERCIO	25. TRANSPORTE Y ALMACENAMIENTO	26. COMUNICACIONES	27. SERVICIOS FINANCIEROS	28. SERVICIOS A LAS EMPRESAS	29. PROPIEDAD DE VIVIENDA	30. SERVICIOS COMUNALES, SOCIALES Y PERSONALES	31. RESTAURANTES Y HOTELES	32. SERVICIOS DOMÉSTICOS	33. SERVICIOS DE LA ADMINISTRACIÓN PÚBLICA	34. IMPUTACIÓN BANCARIA	35. INTERMEDIO POR PRODUCTOS FINANCIEROS
PRODUCTOS		2,984,747	0	0	1,132,939	0	0	0	111,272	0	2,981,987	63,947	188,412	431,063	0	521	0	1,897	0	0	0	0	0	0	0	0	0	0	0	0	192,227	545,208	0	657,931	0	8,872,129
1. PRODUCTOS AGRÍCOLAS NO INDUSTRIALES		2,984,747	0	0	1,132,939	0	0	0	111,272	0	2,981,987	63,947	188,412	431,063	0	521	0	1,897	0	0	0	0	0	0	0	0	0	0	0	192,227	545,208	0	657,931	0	8,872,129	
2. PRODUCTOS AGRÍCOLAS INDUSTRIALES		0	1,598,386	0	96,115	0	0	0	0	0	0	868,418	6,152,684	41,915	7,035	55,704	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8,823,259	
3. COCA		0	0	17,142	0	0	0	0	0	0	0	0	0	0	0	0	0	111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	17,253	
4. PRODUCTOS PECUARIOS		0	0	0	152,416	0	0	0	6,849,006	543,909	33,736	0	92,714	0	0	19,425	0	1,491	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	17,253	
5. SILVICULTURA, CAZA Y PESCA		181,886	60,007	0	18,546	58,324	7,012	593,179	385,961	0	51,768	0	0	0	18,546	994,488	50,569	21,257	0	31,447	0	0	0	545,089	0	0	0	0	0	0	0	0	0	0	17,253	
6. PETRÓLEO CRUDO Y GAS NATURAL		0	0	0	0	0	7,048,788	0	0	0	0	526,788	536,047	371,863	0	60,166	0	26,358	4,544,295	2,291,670	0	0	1,483,162	0	0	0	0	0	0	0	0	0	0	0	0	16,853,172
7. MINERALES METÁLICOS Y NO METÁLICOS		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13,967	1,112,378	2,480,038	0	77,483	0	1,831,234	0	0	0	0	0	0	0	0	0	0	0	1,537,860
8. CARNES FRESCAS Y ELABORADAS		0	0	0	0	0	0	0	1,424,691	37,044	0	86,688	0	0	206,318	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3,228,912	
9. PRODUCTOS LÁCTEOS		0	0	0	0	0	0	0	0	147,303	19,655	0	0	172,258	27,888	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	506,712	827,311	0	142,156	0	788,972
10. PRODUCTOS DE PANADERÍA Y PAÑALES		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3,971,935		
11. AZÚCAR Y CONFITERÍA		0	0	0	0	0	0	0	0	201,228	53,788	125,287	61,759	629,953	84	0	0	1,083	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1,306,800	
12. PRODUCTOS ALIMENTICIOS DIVERSOS		0	0	0	1,475,207	12,312	0	510,383	115,473	73,626	75,478	6,931	362,281	0	0	0	0	87,885	0	0	0	0														

Source: Prepared by the author based on data from the National Institute of Statistics of Bolivia.

The following section presents the calculation of the inverse matrix applied to the difference between the identity matrix and the matrix of technical coefficients, which results from weighting the values of each cell in the matrix above by the production values of each sector for the Leontief calculations.

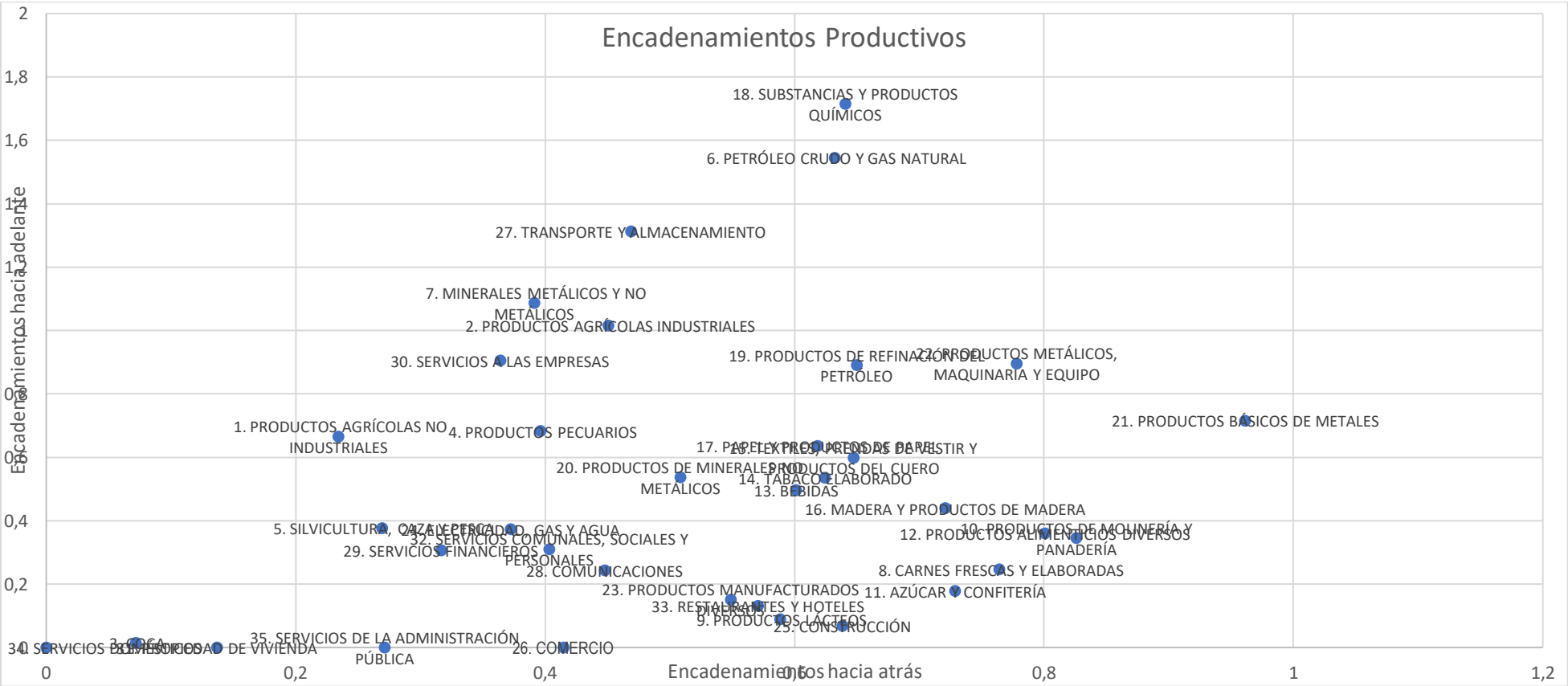
Table 3. Matrix for multiplier calculations

[illegible]

Source: Prepared by the author based on data from the National Institute of Statistics of Bolivia.

The results of the forward and backward multipliers are shown in the following figure.

Figure 14. Productive linkages



Source: Prepared by the author based on INE data.

As shown in the graph, sectors such as basic metal products, tanning, wood, construction, hotel services, and the food industry in general have a strong effect in integrating other productive sectors. At the same time, sectors such as chemical products, crude oil and natural gas, transport, industrial agricultural products, machinery, and equipment occupy an important position in driving production in other sectors. It should be emphasized that the chemical products industry, machinery and equipment (still to be developed in the country), and petroleum refining play an important role both in integrating and in driving productive sectors.

Importance of Investment

According to economic theory, production is directly related to the capital stock, which evolves over time through investment and depreciation, the population and its corresponding growth rate, and factors linked to technology. The base model corresponds to the Cobb-Douglas model, shown as follows:

$$Producción = A * K^{\alpha} * Población^{1-\alpha}$$

Where A is the constant that mainly includes technological development, and K is the capital stock.

A Cobb-Douglas-type model is not easy to establish for the Bolivian economy, because the capital stock must first be calculated and then econometric models must be applied to determine the factors and parameters of the indicated equation.

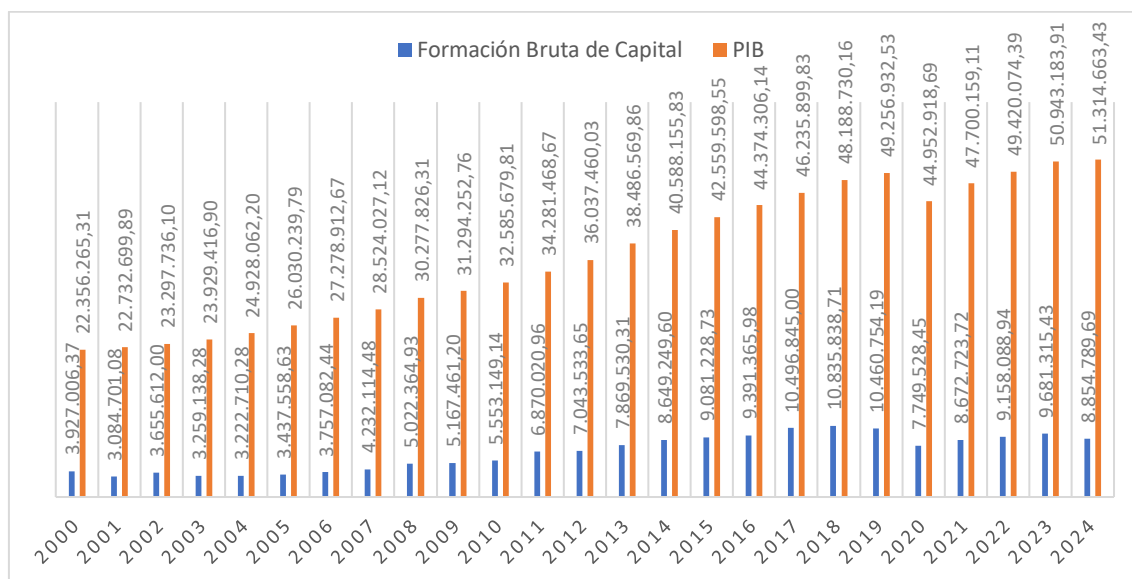
The relationship between investment and the capital stock can be represented simply as follows:

$$K_t = K_{t-1} + I_t - d * K_{t-1}$$

Where I is investment, which for statistical data purposes may be approximated by gross capital formation, and d corresponds to the capital depreciation factor, which could be considered constant.

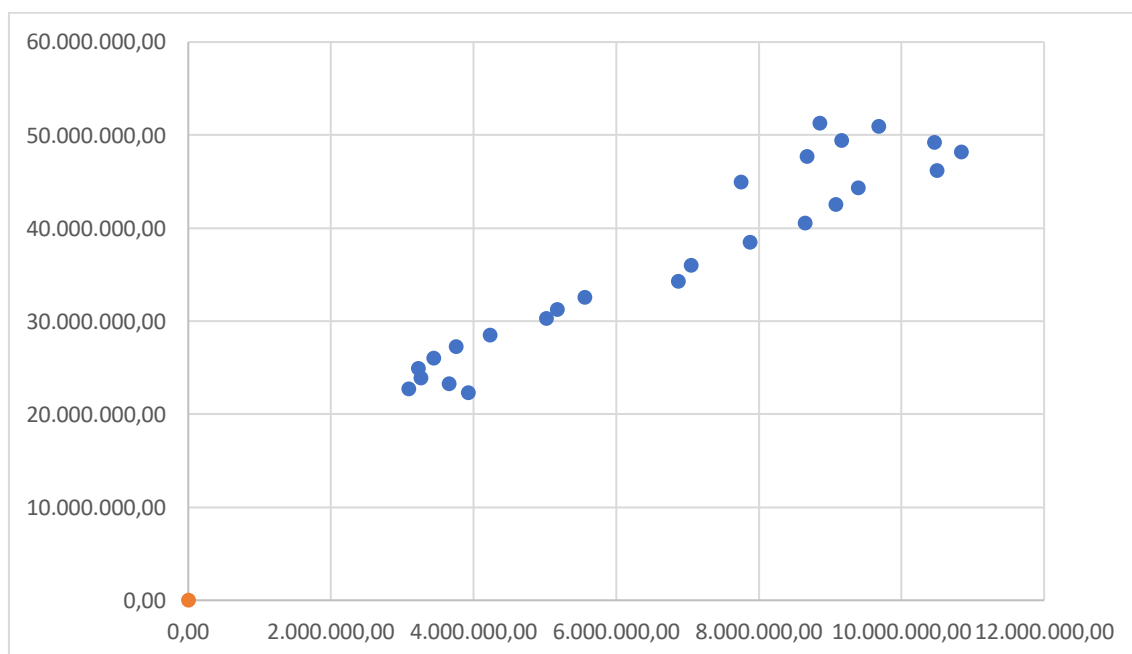
The following figure shows the evolution of gross capital formation and Bolivia's GDP at constant 1990 prices.

Figure 15. Evolution of GDP and gross capital formation at constant 1990 prices for the Bolivian economy



Source: Prepared by the author based on INE data.

Figure 16. Evolution of GDP and gross capital formation at constant 1990 prices for the Bolivian economy, scatter plot



The scatter plot shows a linear relationship; applying a regression yields the following parameters and results.

Table 2. Regression parameters between gross capital formation and GDP.

<i>Regression statistics</i>	
Multiple correlation coefficient	0,957725978
Coefficient of determination R ²	0,917239048
R ² ajustado	0,913640746
Standard error	3002144,043
Observaciones	25

	<i>Coefficients</i>	<i>Standard error</i>	<i>t-statistic</i>	<i>Probability</i>
Intercept	12132792,07	1651906,5	7,34472082	1,7994E-07
Variable X 1	3,631774117	0,22747118	15,9658648	6,1455E-14

The parameters show a consistent equation that could be used for national production optimization models.

Taking into account the relationship between investment and national production, and considering different levels of investment that would increase GDP, while recognizing that on the demand side these must increase by the same magnitude, the following model is proposed.

$$Max = \Delta Inv_i * (I - A)^{-1}$$

Sujeto a:

$$\sum_{i=1}^N Inv_i * f_i \leq B$$

Where:

Inv_i = Inversión para cada sector

I = matriz identidad

A = matriz de coeficientes técnicos, insumo producto

f_i = factor de inversión

B = presupuesto de inversión

Considering that the objective is to maximize the highest gross production value by taking into account the strongest linkage with sectors according to the Leontief multipliers, which use the matrix of technical coefficients from the input-output matrix A, with the aim of linking the largest possible number of sectors.

Conclusions and Recommendations

- Industrial production in Bolivia is underdeveloped and, in general, has a low level of value added and limited incorporation of technology.
- Labor productivity, according to studies carried out in this field, shows levels below the Latin American average.
- Manufacturing industrial production in Bolivia is highly dependent on imports of inputs and processed manufactures, and it lacks a connected fabric among the different sectors, except for the linkages related to agricultural and food production.
- The level of logistics development is low, both in terms of infrastructure and information systems, and costs related to logistics and transport are significant and higher than those of countries in the region.
- A multiplier analysis shows that sectors such as basic metal products, tanning, wood, construction, hotel services, and the food industry in general have a strong effect in integrating other productive sectors. At the same time, sectors such as chemical products, crude oil and natural gas, transport, industrial agricultural products, machinery, and equipment occupy an important position in driving production in other sectors. It should be emphasized that the chemical products industry, machinery and equipment (still to be developed in the country), and petroleum refining play an important role both in integrating and in driving productive sectors.

The path to strengthening the industrial sector involves:

- Developing industries for the production of inputs and processed raw materials mainly for industry (steelmaking, chemicals, industrial equipment, polymers, and the lithium industry), thereby strengthening the domestic industrial fabric and allowing the opening of foreign investment through an aggressive and creative tax policy that generates significant advantages.
- Strengthening technical and technological education by generating partnerships among industry, the state, and academia, while enabling the integration of international support through cooperation and association.
- Increasing manufacturing exports by improving quality, incorporating greater value added and technology, and developing new industries such as software and tourism, as well as mining and hydrocarbons by increasing exploration while considering their industrialization within the country as far as possible. This must also be accompanied by large-scale logistics development and an appropriate strategy in the area of international relations for the opening and consolidation of markets.

- Improving the technical and educational level by strengthening and increasing human capital.
- Improving the country's logistics conditions in terms of both infrastructure and information technologies, enabling better international integration and allowing Bolivia to take advantage of its geographical position in the center of South America.
- Developing renewable energy generation systems that allow the gradual replacement of fossil fuels and the corresponding dependence on them.

Bibliography

- National Logistics Survey, Ministry of Productive Development and Plural Economy, 2019.
- National Institute of Statistics, information on national accounts.
- Labor Productivity, INESAD, Beatriz Muriel, Carmen Gabriela Olivares.